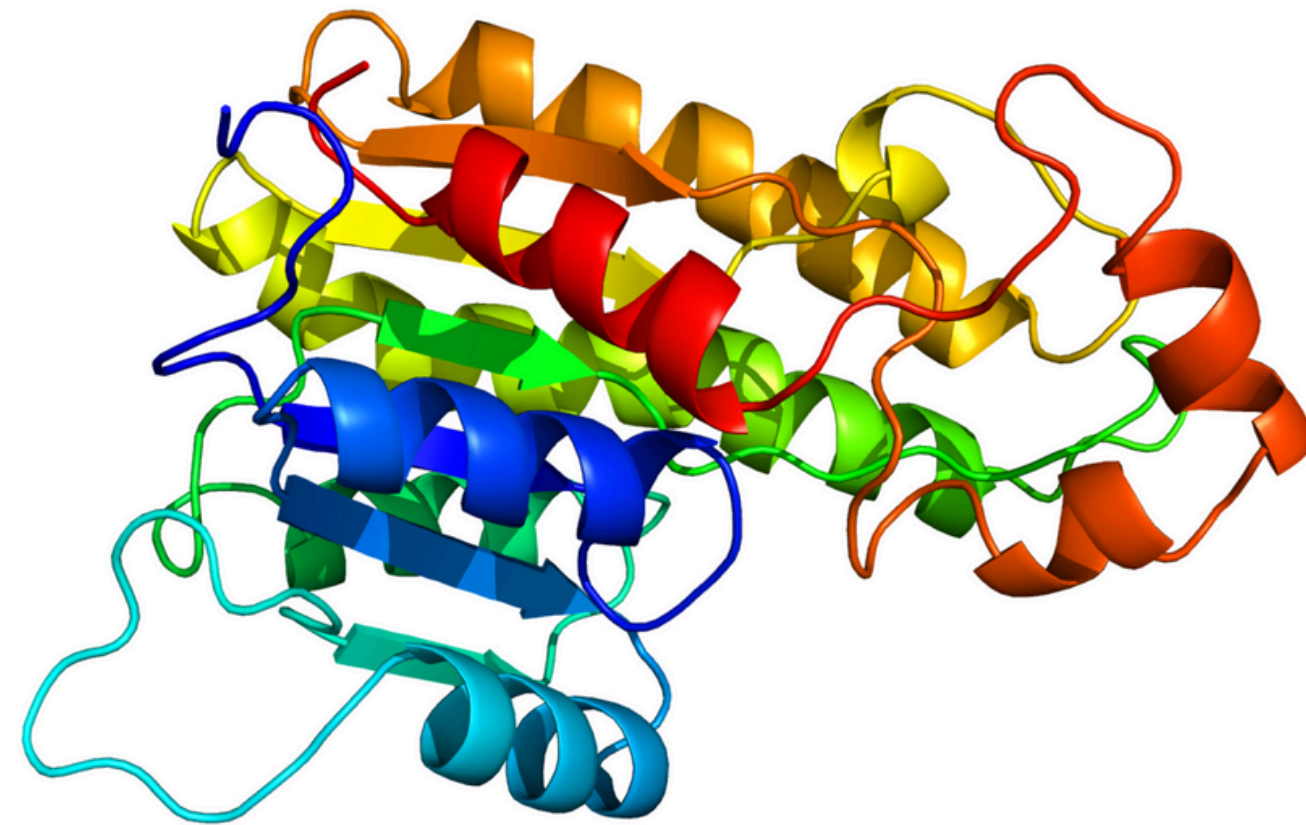


# ALPHA FOLD 3 and ESMfold FROM YOUR TERMINAL

## AlphaFoldER

Michał Zgieb  
Kacper Pietrzyk  
Katsiaryna Dubrouskaya  
Veranika Kananovich

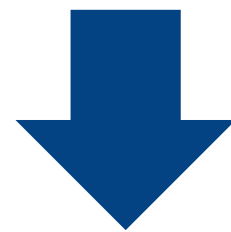


Architecture of large bioinformatics projects, 2025

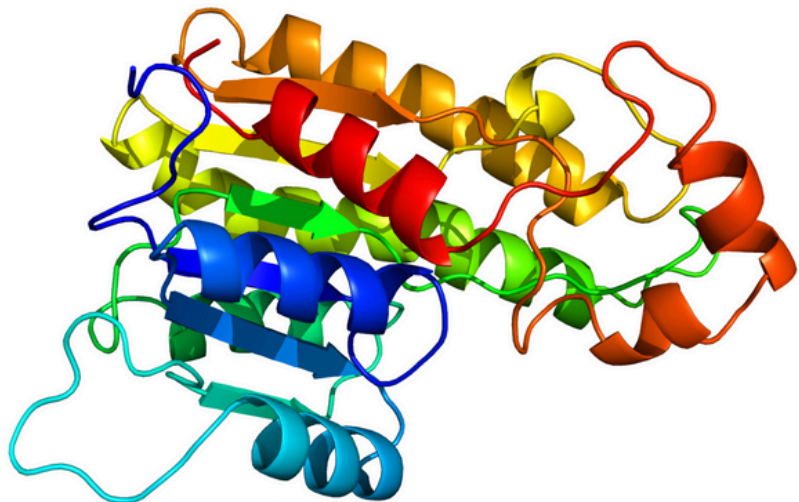
# ESM FOLD



MSEQTKDDDISFEANNTEMMMTFQIQRIYTKDISLDLTTTKDIS



ESM Fold API



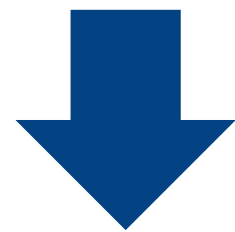
ESMFold provides an API for accessing their server locally. This step can be integrated into other parts of your code and is very simple to use.

However, being a smaller language model, ESMFold might not provide the most accurate predictions. A better option, of course, would be to use AlphaFold 3....

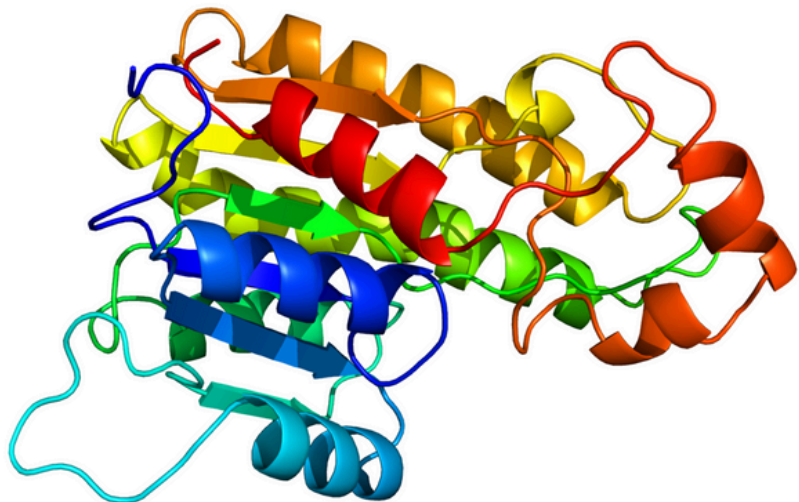
# ALPHA FOLD



MSEQTKDDDISFEANNTEMMMTFQIQRIYTKDISLDLTTTKDIS



no Alfafold Fold API

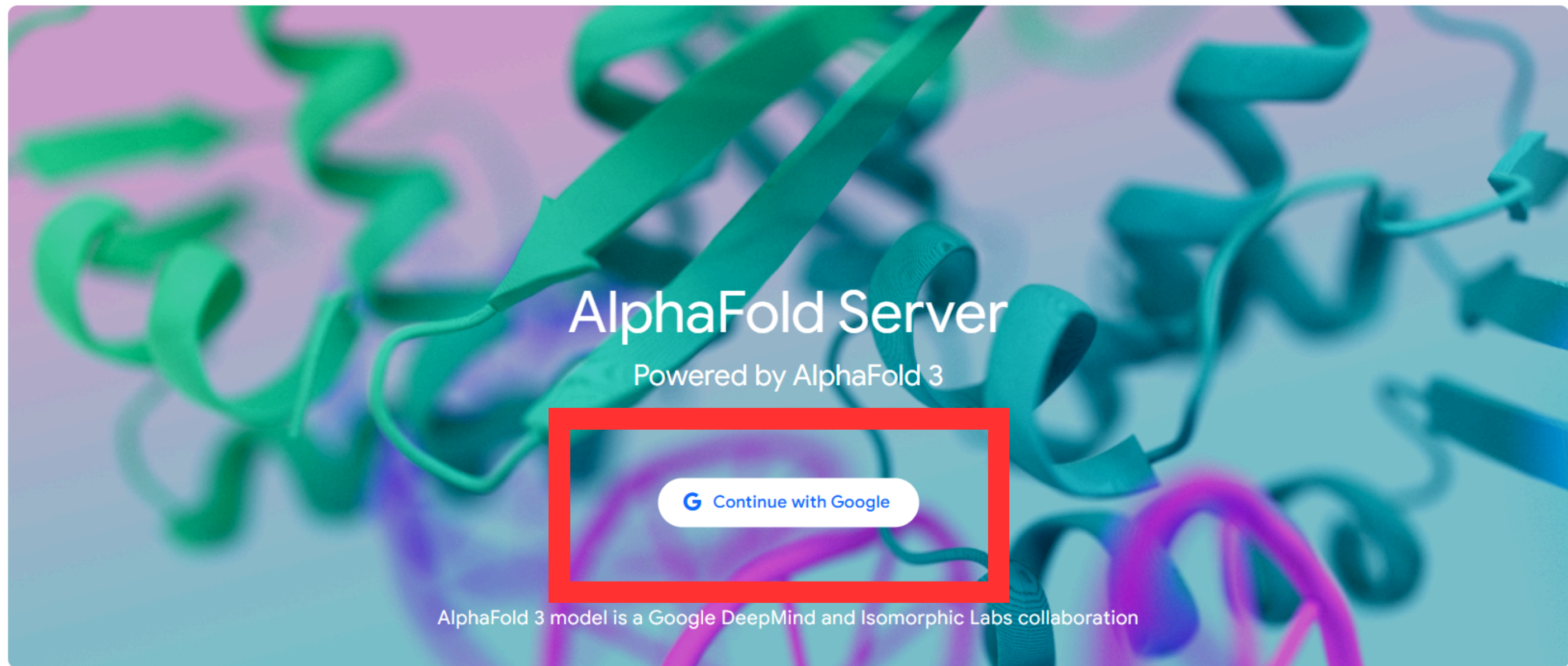
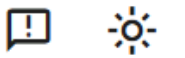


AlphaFold 3 is the newest AlphaFold model for predicting protein structure...  
However there is no API provided by the company and the only way to access the server is through the website....



# HOW IT IS NORMALLY DONE

AlphaFold Server   Server   About   FAQ & Guides ▼



# HOW IT IS NORMALLY DONE

AlphaFold Server

Server

About

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Remaining jobs: 30

AlphaFold Server allows you to model a structure consisting of many biological molecules

[Learn more](#) ^

- Remaining jobs refresh each day
- Jobs can be up to 5,000 tokens - see more details on token calculation, accepted formats, seed selection and other features in our [FAQ](#)
-  Use the entity bar to chemically modify proteins and nucleic acids
-  Get in touch with the AlphaFold team if you have any questions

Explore these examples of structures to see it in action – try them out without using your quota until you begin editing!

 Protein-RNA-Ion: PDB 8AW3

 Protein-Glycan-Ion: PDB 7BBV

 Protein-DNA-Ion: PDB 7RCE

Ok, got it

# HOW IS IT NORMALLY DONE



Entity type  
Protein ▾

Copies  
1

>Paste sequence or fasta  
Input

▾

+ Add entity

Save job

Continue and preview job

| Search History                                                                                                     |                    |                  |   |
|--------------------------------------------------------------------------------------------------------------------|--------------------|------------------|---|
| <div>✓ Completed</div> <div>✓ Saved draft</div> <div>✓ In progress</div> <div>✓ Examples</div> <div>✓ Failed</div> |                    |                  |   |
| <input type="checkbox"/>                                                                                           | Name               | Modified         |   |
| <input type="checkbox"/>                                                                                           | ✓ 2025-06-09_00:26 | 2025-06-09 00:26 | ⋮ |
| <input type="checkbox"/>                                                                                           | ✓ 2025-06-09_00:09 | 2025-06-09 00:10 | ⋮ |

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[Continue and preview job](#)

 Search History

[✓ Completed](#) [✓ Saved draft](#) [✓ In progress](#) [✓ Examples](#) [✓ Failed](#)

| <input type="checkbox"/> | Name                                                 | Modified       |
|--------------------------|------------------------------------------------------|----------------|
| <input type="checkbox"/> | <input checked="" type="checkbox"/> 2025-06-09_00:26 | 25-06-09 00:26 |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> 2025-06-09_00:09 | 25-06-09 00:09 |

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# WHAT CAN BE DONE INSTEAD



All-in-one Browser Automation Framework:  
Web Crawling / Testing / Scraping / Stealth

pypi v4.39.2 release v4.39.2 docs seleniumbase.io CI build passing Discord 156 online

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[GUI](#) | [TestPage](#) | [UC Mode](#) | [CDP Mode](#) | [Charts](#) | [Grid](#)  
[How](#) | [Migrate](#) | [CasePlans](#) | [Template](#) | [Hybrid](#) | [Tours](#)  
[CI/CD](#) | [JSMgr](#) | [Translator](#) | [Presenter](#) | [Dialog](#) | [Visual](#)

**Selenium WebDriver** is a popular open-source tool used for automating web application testing. It allows developers and testers to programmatically control a web browser's behavior, simulating user interactions like clicking buttons, filling out forms, navigating pages, etc

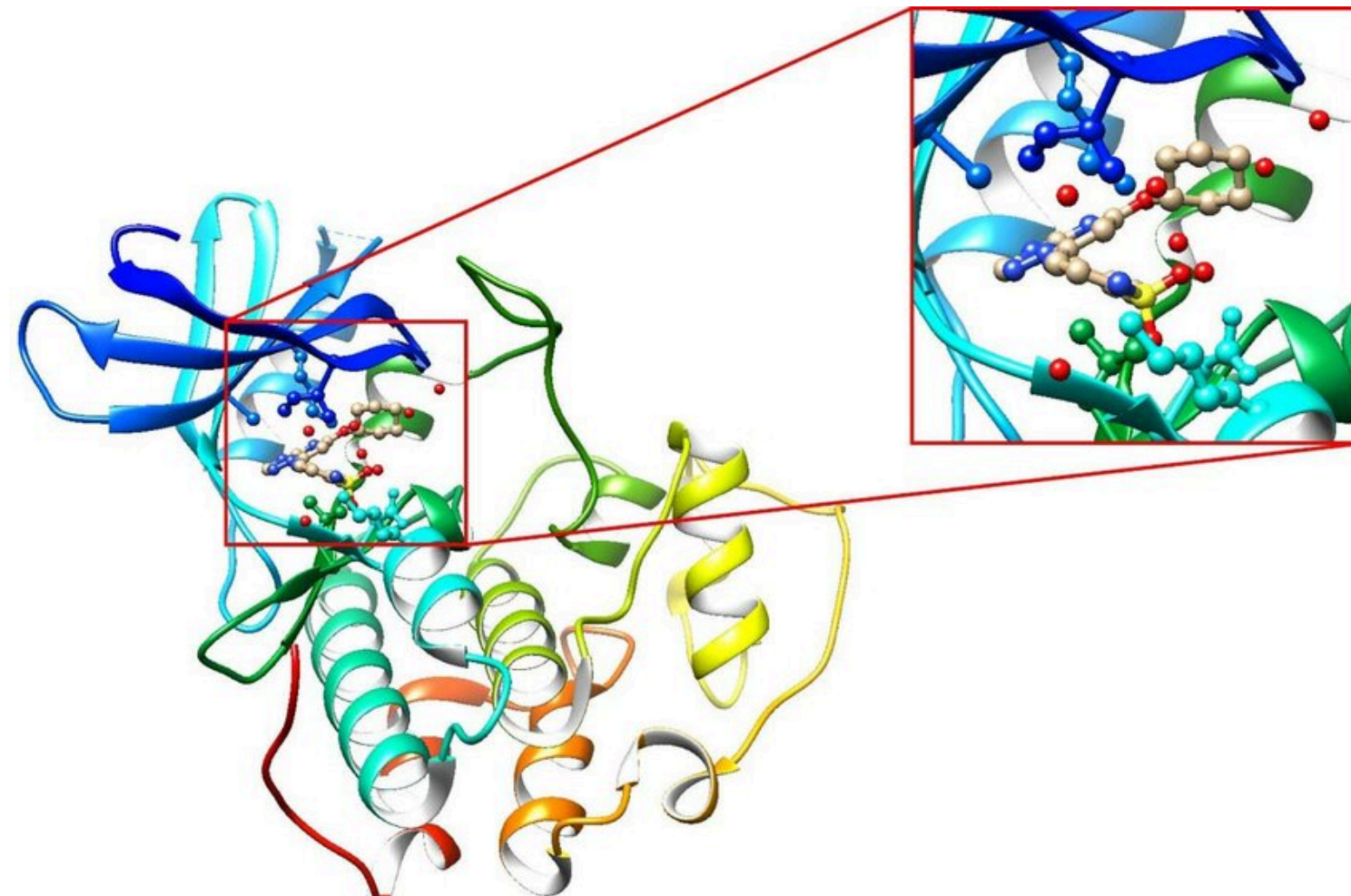


# ALPHAFOLDER WORKFLOW

1. Launch the Google Chrome browser on a specified port to be used later by Selenium.
2. Allow the user to manually log in to their Google account, as automated login might be blocked.
3. Open the AlphaFold server web page.
4. Parse a FASTA file, treating each sequence's header as a job name.
5. Submit each sequence as a separate job on the AlphaFold server.
6. Track the number of active jobs, respecting the session's allowed job limit.
7. Download completed jobs once they're finished

# NOTES FOR THE FUTURE:

- Taking into account options like : adding multiple chains, ligands, nucleic acids, ions
- Comparing ESMFold and AlphaFold3 predictions





# THANK YOU!

